TOBB ETÜ
University of Economics & Technology

Data Mining
Project
Instructions & Guidelines

BIL 476 / 573 — Data Mining

Summer 2026

Project Weight: 20% of Final Grade

Duration: 6 Weeks

Type: Individual, Open-Book

Last Updated: June 11, 2026

Contents

1 Project Overview

Summary

Each student will individually select **one core data mining topic** from the syllabus, apply it to a real-world dataset, and produce a professional report in **IEEE conference paper format**. The project accounts for **20% of the final grade**.

1.1 Objectives

The purpose of this project is to:

1. Apply data mining concepts and techniques learned in class to a **real-world problem**.
2. Gain hands-on experience with the **complete data mining pipeline**: data collection, preprocessing, algorithm application, evaluation, and interpretation.
3. Develop skills in **scientific report writing** using professional formats (IEEE conference paper).
4. Practice **critical analysis** — not just applying algorithms, but understanding why they work (or don't) and what the results mean.
5. Compare **multiple approaches or variations** and analyze trade-offs.

1.2 Key Rules

Important Rules

- This is an **individual project**. Each student works alone.
- The project is **open-book**. You may use any resources, but all sources must be properly cited.
- The final report must be submitted as a **PDF** in **IEEE conference paper format**.
- A separate **Personal Reflection & AI Use Disclosure Form** must be submitted alongside the report.
- Any use of AI tools must be **declared clearly**; AI may support learning, coding, debugging, language polishing, or LaTeX formatting, but it must not replace your own thinking, analysis, interpretation, or report writing.
- **Academic integrity:** All work must be your own.

Late Submission Policy

- A penalty of **20% per day** will be applied for late submissions.
- Reports submitted **more than 5 days late will NOT be accepted** (grade of 0).

2 Topic Selection

You must choose **one** data mining topic for your project, aligned with the course syllabus. You may also propose your own topic; please consult within the first week.

2.1 Available Topics

Students may select **one** topic area from the tables below. Topics 1–6 focus on **pattern mining, association rules, and core data mining concepts**, which constitute a major part of the course. Topics 7–14 cover other important areas of data mining.

Part A: Pattern Mining, Association Rules, and Core Data Mining Concepts

#	Topic Area	Example Tasks / Questions
1	**Market Basket Analysis**	Apply association rule mining to retail transaction data (e.g., grocery, e-commerce). Compare Apriori vs. FP-Growth in efficiency and scalability. Evaluate rules using support, confidence, lift, and at least one null-invariant measure (e.g., Kulczynski, cosine). Identify misleading rules from the support-confidence framework.
2	**Advanced Pattern Mining**	Mine **closed frequent itemsets** and **max-patterns** and compare with the full set of frequent itemsets. Analyze compression ratio and information loss. Explore **constraint-based mining** (anti-monotone, monotone, convertible, or succinct constraints) to prune the search space.
3	**Sequential Pattern Mining**	Discover sequential patterns in time-ordered data: customer purchase sequences, web clickstream logs, medical treatment pathways, or course enrollment sequences. Apply GSP, PrefixSpan, or SPADE. Analyze how support thresholds and gap constraints affect discovered patterns.
4	**Correlation Analysis & Rule Evaluation**	Mine association rules and perform an in-depth **interestingness measure comparison**. Evaluate rules using support, confidence, lift, χ^2, all-confidence, max-confidence, Kulczynski, and cosine. Identify cases where measures disagree and analyze why. Discuss the **null-invariance problem**.
5	**Multi-Level & Multi-Dim. Mining**	Explore association rules at **multiple concept levels** using concept hierarchies (e.g., categories $\rightarrow$ subcategories $\rightarrow$ items). Mine **multi-dimensional association rules** combining quantitative and categorical attributes. Discuss redundancy filtering across hierarchy levels.
6	**Data Preprocessing & Its Impact on Mining**	Investigate how different preprocessing strategies affect downstream mining quality. Compare missing value treatments (deletion, imputation, model-based), normalization (min-max, z-score), discretization (equal-width, equal-frequency, clustering-based), and dimensionality reduction (PCA, feature selection) on a mining or classification task.

Part B: Classification, Clustering, and Other Data Mining Topics

#	Topic Area	Example Tasks / Questions
7	**Classification**	Predict disease diagnosis, student dropout, spam detection, or credit risk. Compare Decision Tree, Naïve Bayes, k-NN, and an ensemble method (Random Forest, AdaBoost, or XGBoost). Report accuracy, precision, recall, F1-score, and ROC/AUC. Discuss class imbalance effects.

#	Topic Area	Example Tasks / Questions
8	**Clustering**	Segment customers, group documents, or cluster geographic/sensor data. Apply K-Means, hierarchical clustering, and DBSCAN. Evaluate with silhouette score, SSE, Davies-Bouldin index, and visual analysis. Discuss the effect of different distance measures.
9	**Anomaly / Outlier Detection**	Detect fraudulent transactions, network intrusions, or manufacturing defects. Compare statistical (z-score, IQR), distance-based (k-NN), and density-based (LOF) methods. Discuss false positive/negative trade-offs.
10	**Dimensionality Reduction & Visualization**	Apply PCA, t-SNE, or UMAP to a high-dimensional dataset. Analyze the effect on downstream classification or clustering accuracy. Visualize and interpret reduced representations. Compare linear vs. nonlinear methods.
11	**Recommendation Systems**	Build a collaborative filtering or content-based recommendation system using transaction or rating data. Evaluate using RMSE, precision@k, recall@k, or NDCG. Discuss cold-start and sparsity challenges.
12	**Sequence / Time Series Analysis**	Predict stock prices, classify ECG signals, forecast energy consumption, or detect anomalies in sensor streams. Compare traditional methods (ARIMA, sliding window) with deep learning (LSTM, Transformer).
13	**Deep Learning for Data Mining**	Apply MLP, CNN, RNN, or Transformers to a classification, clustering, or prediction task. Compare with traditional data mining methods. Discuss trade-offs: accuracy vs. interpretability, training cost, data requirements.
14	**Comparative Study**	Choose one dataset and rigorously compare 3+ algorithms within a single task (e.g., 4 classifiers, 3 clustering methods, or multiple pattern mining approaches with different parameters). Provide statistical significance analysis.

2.2 Sample Starter Questions

To help you get started, here are sample example questions you could investigate:

Q1. *"What are the most frequent item combinations purchased together in grocery data, and which association rules are truly interesting beyond simple support-confidence?"*

Q2. *"Can we predict heart disease from patient records? How do different classifiers compare, and what is the effect of feature selection?"*

Q3. *"Are there natural groupings among countries based on socioeconomic indicators? How do K-Means, DBSCAN, and hierarchical clustering compare?"*

Q4. *"Can we identify unusual network traffic patterns indicating cyber attacks? Which outlier detection method is most effective?"*

Q5. *"How does PCA vs. t-SNE vs. UMAP affect the visualization and clustering quality of a high-dimensional text dataset?"*

Q6. *"Can a deep neural network outperform traditional classifiers on tabular healthcare data? Under what conditions?"*

Q7. *"What is the effect of different data transformations (log, z-score, min-max) on clustering quality for a given dataset?"*

Q8. *"Can sequential pattern mining reveal meaningful treatment pathways in medical records data?"*

Tip

You are encouraged to formulate your **own research question**. The best projects are those driven by genuine curiosity about a problem. If you have an idea that does not fit neatly into the categories above, discuss it in the early stages.

2.3 Kaggle Competition Option

You may choose to participate in a **Kaggle competition** as your course project:

- Select an active or recently completed competition from `https://www.kaggle.com/competitions`
- Follow the competition's problem definition and dataset
- Your report must still follow the required structure (Section 7)
- Include your Kaggle leaderboard score/ranking in the results section

Graduate Students (BIL 573)

The Kaggle competition option is **especially recommended** for graduate students. It provides a competitive benchmark and real-world evaluation framework.

3 Differentiated Expectations

3.1 Undergraduate Students (BIL 476)

- Apply **standard methods** from the course correctly and thoroughly.
- Compare at least **2–3 different approaches** and analyze results.
- Demonstrate solid understanding of preprocessing, algorithm application, and evaluation.
- Literature review: at least **3–5 references**.

3.2 Graduate Students (BIL 573)

Additional Expectations for Graduate Students

In addition to the undergraduate expectations, graduate students are expected to:

- **Go beyond standard methods.** Explore:
  - **Hybrid approaches** (combining two or more techniques)
  - **Novel structures or modifications** to existing algorithms
  - **Advanced techniques** not covered in class (with proper explanation)
- Produce work that could potentially lead to an **academic publication**.
- Use **LaTeX (IEEE format)** for the report (strongly recommended).
- Provide a deeper literature review: at least **5–8 references**.
- Demonstrate more rigorous experimental methodology and analysis.

These factors will be taken into account in the grading of the project.

4 Data Sources

You may obtain your dataset from any of the following sources:

Table 3: Recommended Data Sources

Source	**URL / Description**
Kaggle	`https://www.kaggle.com/datasets` — Thousands of curated datasets with community notebooks
UCI ML Repository	`https://archive.ics.uci.edu` — Classic machine learning and data mining datasets
GitHub	Various repositories with open datasets
Data.gov	`https://www.data.gov` — US government open data
Open Data Portals	Country/city open data portals (e.g., EU Open Data, Turkish open data portals)
Web Scraping	You may scrape data from websites. Ethical considerations must be documented in your report.
Synthetic Data	Quality synthetic data generation using tools like `sklearn.datasets`, `Faker`, or generative models. You must justify why synthetic data is appropriate.

Minimum Data Requirements

- Dataset should have **at least 1,000 records**.
- Dataset should have **at least 8 attributes**.
- Mixed attribute types (nominal, ordinal, numeric) are preferred.

5 Tools and Programming Languages

5.1 Programming Languages

Projects must be implemented in one of the following languages:

- **Python** (recommended)
- **Java**
- **MATLAB**
- **C / C++**
- **R**

5.2 Recommended Python Libraries

5.3 Development Environments

- Jupyter Notebook / JupyterLab
- Google Colab (`https://colab.research.google.com`)
- VS Code, PyCharm, or any IDE of your choice

Table 4: Recommended Python Libraries

Library	Purpose
`pandas`	Data manipulation and analysis
`numpy`	Numerical computing
`scikit-learn`	Machine learning algorithms, preprocessing, evaluation
`mlxtend`	Association rule mining (Apriori, FP-Growth)
`matplotlib` / `seaborn`	Data visualization
`scipy`	Statistical analysis
`tensorflow` / `pytorch`	Deep learning (if applicable)
`networkx`	Graph/network analysis (if applicable)

6 Report Format

6.1 Final Report

The final report **must be submitted as a PDF** in **IEEE Conference Paper format**.

- IEEE template download: `https://www.ieee.org/conferences/publishing/templates.html`
- A customized LaTeX template is provided separately with this assignment.
- The report must include the AI Assistance Declaration described in Section 10.

Table 5: Report Preparation Options

Format	Tool	Notes
LaTeX *(Strongly recommended for BIL 573)*	Overleaf	Online collaborative LaTeX editor. Free tier available. `https://www.overleaf.com`
	Prism	AI-assisted LaTeX editor with OpenAI integration. Use only within the AI policy in Section 10. `https://prism.openai.com`
	TeXstudio, MiKTeX, TeX Live	Local LaTeX installations.
Word	Microsoft Word	Use the IEEE Word template. Export/save as PDF.
	LibreOffice Writer	Free alternative. Export as PDF.
	Google Docs	Export as PDF. May require manual formatting.
Jupyter Notebook	Jupyter Notebook / Lab	Can be used **alongside** the main report to show step-by-step code and analysis. Export as PDF.
	Google Colab	If used as the primary format, must include full markdown explanations interleaved with code.

7 Required Report Structure

Regardless of the tool used, the report must contain the following sections:

I. Title, Author, Abstract

- Project title

- Student name, student ID, email
- Abstract: 150–250 words summarizing the problem, approach, key results, and conclusions
- Keywords: 4–5 relevant terms

II. Introduction

- Problem definition and motivation
- Research question or objective
- Brief overview of approach
- Paper structure outline

III. Related Work

- Literature review (BIL 476: 3–5 refs; BIL 573: 5–8 refs)
- How your work relates to or differs from existing studies

IV. Dataset Description

- Data source and URL
- Data characteristics (records, attributes, types, target variable)
- Descriptive statistics
- Ethical considerations

V. Methodology

- Data preprocessing steps with justification
- Algorithm(s) chosen with theoretical background and justification
- Multiple approaches/variations tried
- Experimental setup (split, cross-validation, hyperparameters, tools)

VI. Results

- Evaluation metrics (appropriate to the task)
- Results tables and figures
- Visualizations (confusion matrices, ROC curves, cluster plots, etc.)
- Comparison of approaches

VII. Discussion

- Which method performed best and why
- Limitations of the approach
- Surprising or unexpected findings
- Effect of data quality issues
- Ethical implications
- (BIL 573) Novelty and publication potential

VIII. Conclusion and Future Work

- Summary of key findings
- Lessons learned
- Concrete future work directions

IX. AI Assistance Declaration

- Required in the report, placed before References or in an Acknowledgment/Declaration section.

- If no AI tools were used, state that no generative AI or AI-assisted tools were used for the submitted report, code, analysis, figures, or interpretation.
- If AI tools were used, name the tool(s), describe the purpose of use, identify the affected parts of the work, and briefly explain how the outputs were verified.
- AI tools must not be listed as authors, sources of factual claims, or substitutes for the student's own analysis.

X. References

- IEEE citation format
- Numbered consecutively within brackets

XI. Appendix (optional)

- Code repository link
- Key code snippets
- Additional figures and tables

8 Grading Rubric

Table 6: Grading Rubric

Component	**BIL 476**	**BIL 573**	**Description**
Problem Definition & Motivation	10%	8%	Clear, well-motivated problem statement with a specific research question
Data Understanding & Preprocessing	15%	12%	Thorough EDA, appropriate and justified preprocessing steps
Methodology & Algorithm Selection	20%	18%	Correct application, justified choices, multiple approaches compared
Results & Evaluation	25%	22%	Proper metrics, clear visualizations, rigorous comparisons
Critical Analysis & Discussion	15%	15%	Depth of independent analysis, limitations acknowledged, ethical considerations
Report Quality & Formatting	10%	10%	Professional IEEE format, proper citations, clear writing, appropriate AI-use declaration
Code Quality & Reproducibility	5%	5%	Clean, documented, reproducible code with clear instructions and verified AI-assisted code, if any
Novelty & Advanced Approaches	—	**10%**	Hybrid methods, novel modifications, publication potential *(BIL 573 only)*
Total	**100%**	**100%**	

8.1 Grading Scale Guidelines

9 Submission Checklist

Before submitting, ensure you have completed all of the following:

Table 7: Quality Indicators by Grade Range

Grade Range	Description
90–100	Exceptional work. Novel insights, rigorous methodology, publication-quality report, thorough analysis, excellent visualizations.
80–89	Very good work. Solid methodology, good comparisons, well-written report, meaningful analysis with minor gaps.
70–79	Good work. Correct application of methods, adequate comparisons, acceptable report quality, some analysis depth missing.
60–69	Satisfactory work. Basic application of methods, limited comparisons, report needs improvement, shallow analysis.
Below 60	Insufficient work. Incorrect methodology, missing components, poor report quality, lack of analysis, or extensive AI-generated content that does not demonstrate the student's own understanding.

Table 8: Final Submission Checklist

#	**Item**	**Check**
1	Main project report (PDF, IEEE conference format)	☐
2	Personal Reflection & AI Use Disclosure Form (separate PDF)	☐
3	AI Assistance Declaration included in the report, even if no AI tools were used	☐
4	Source code (GitHub link, Google Colab link, or ZIP file)	☐
5	Jupyter Notebook with step-by-step explanation (if applicable)	☐
6	Dataset included or clearly linked/referenced	☐
7	All figures and tables are properly labeled and referenced	☐
8	All references are in IEEE format and properly cited	☐
9	Abstract is 150–250 words, no symbols or special characters	☐
10	Report contains all required sections (Section 7)	☐
11	Code is clean, documented, and reproducible	☐
12	The report reflects your own ideas, analysis, interpretation, and writing	☐
13	All template/guidance text removed from the report	☐

10 Academic Integrity

Academic Integrity Policy

1. This project is to be completed as **individual work**.
2. All sources, references, datasets, code libraries, and tools must be **properly cited or declared**.
3. **Plagiarism** (copying code, text, figures, or results from others without appropriate attribution) is inconsistent with academic integrity principles.
4. **Fabrication or falsification** of results, datasets, experiments, citations, or performance claims is not acceptable.
5. **AI tool usage** (e.g., ChatGPT, GitHub Copilot, Claude, Gemini, Grammarly, or AI-assisted IDEs) is permitted only as a supporting tool and must be **clearly and transparently reported** in both the report's AI Assistance Declaration and the Personal Reflection & AI Use Disclosure Form.
6. You remain fully responsible for the accuracy, originality, legality, reproducibility, and integrity of all submitted work, including any AI-assisted text, code, analysis, or figures.
7. Failure to adhere to these principles may affect the evaluation of the project in accordance with university regulations.

10.1 Responsible Use of AI Tools

Permitted Support

AI tools may be used to support learning and project development in limited and responsible ways, such as clarifying concepts, suggesting possible debugging strategies, reviewing code for errors, explaining library functions, improving grammar or readability, fixing LaTeX formatting issues, or helping you understand an algorithm. These uses do not replace your responsibility to understand, verify, and explain every part of the submitted work.

Not Acceptable

The final report must reflect your own ideas, decisions, analysis, interpretation, and writing. It is not acceptable to submit a report, discussion, conclusion, literature review, experimental interpretation, or research argument that is mostly generated by AI and then only lightly edited by the student. AI tools must not be treated as an author, a source of factual claims, or a substitute for reading the original references. AI-generated references, numerical results, explanations, or claims must be independently verified before they are used.

Effect on Grading

Transparent disclosure of permitted AI assistance will not be penalized by itself. However, a large portion of AI-generated report content, weak evidence of the student's own understanding, inability to explain submitted code or results, unverifiable AI-generated claims, or undisclosed AI use may negatively affect the grade. In doubtful cases, the instructor may ask the student to explain the work orally, reproduce parts of the analysis, or provide a brief record of AI prompts and outputs used during the project.

10.2 AI Assistance Declaration Required in the Report

Every report must include a short AI Assistance Declaration before the References section or in an Acknowledgment/Declaration section. Use one of the following formats and adapt it truthfully.

Example Declaration: No AI Used

No generative AI or AI-assisted tools were used to generate the submitted report text, code, analysis, figures, interpretations, or references.

Example Declaration: Limited AI Support Used

AI-assisted tools were used only for limited support, such as code debugging, language polishing, concept clarification, and/or LaTeX formatting. The project idea, methodology, experiments, analysis, interpretation of results, and final report wording were reviewed, verified, and finalized by the student, who accepts full responsibility for the submitted work.

AI Use Disclosure Levels

In the Personal Reflection & AI Use Disclosure Form, students must report AI use using the following compact scale: **0** = not used; **1** = minimal hints; **2** = low support with student-authored final content; **3** = regular support for specific sub-tasks with critical student review; **4** = high support for critical sub-tasks; **5** = very high AI contribution; **6** = AI-led work. Levels **5–6** indicate that the work may fall outside acceptable use for this project and will require careful review.

11 Frequently Asked Questions (FAQ)

Q1: Can I work in a team?
This is an **individual project**. You may discuss ideas and general approaches with classmates; however, each student must independently complete and submit their own report, code, and reflection form. All submitted work must reflect your individual understanding and effort.

Q2: Can I choose a topic not listed in the topic table?
Yes. You may propose a topic related to data mining. Early consultation within the first week is strongly encouraged to help refine and align the topic.

Q3: Can I use a dataset I found on my own?
Yes. You may use datasets from Kaggle, UCI, GitHub, data.gov, open data portals, web scraping, or even generate quality synthetic data. The dataset must meet the minimum requirements (at least 1,000 records and 8 attributes). Document the source clearly in your report.

Q4: Can I participate in a Kaggle competition as my project?
Yes. Select an active or recently completed competition. Your report must still follow the required IEEE format and structure. Include your leaderboard score/ranking. This option is especially recommended for BIL 573 (graduate) students.

Q5: What programming language should I use?
You may use **Python** (recommended), **Java**, **MATLAB**, **C/C++**, or **R**. Choose the language you are most comfortable with and that has adequate library support for your chosen task.

Q6: Do I have to use LaTeX for the report?
No, but it is **strongly recommended**, especially for BIL 573 (graduate) students. You may also use Microsoft Word, LibreOffice, Google Docs, or Jupyter Notebook. Regardless of the tool, the final report must be in **PDF format** following the **IEEE conference paper template**.

Q7: Where can I get the IEEE template?

- **LaTeX:** A customized template is provided with this assignment. You can also download the official template from `https://www.ieee.org/conferences/publishing/templates.html`
- **Word:** Download the IEEE Word template from the same URL above.
- **Overleaf:** Search for "IEEE Conference Template" in the Overleaf template gallery.

Q8: Where can I write LaTeX?
You have several options:

- **Overleaf** (`https://www.overleaf.com`) — Free online LaTeX editor with real-time collaboration.
- **Prism** (`https://prism.openai.com`) — AI-assisted LaTeX editor. Use only for support permitted under the AI policy.
- **Local installation:** TeXstudio, TeXmaker, VS Code with LaTeX Workshop extension, MiKTeX (Windows), TeX Live (Linux/Mac).

Q9: How long should the report be?
There is no strict page limit, but a typical good report is **6–10 pages** in IEEE two-column format (excluding references and appendix). Quality matters more than quantity. Ensure all required sections are covered thoroughly.

Q10: How many algorithms/methods do I need to compare?
You must apply **at least two different approaches or variations** and compare them. This could be:

- Different algorithms for the same task (e.g., K-Means vs. DBSCAN)
- Same algorithm with different parameters or preprocessing
- Standard method vs. a hybrid/novel approach (especially for BIL 573)

Q11: Can I use AI tools like ChatGPT, GitHub Copilot, Claude, Gemini, or Grammarly?
Yes, but only as **supporting tools**. Permitted uses include concept clarification, debugging, code review, limited writing improvement, and LaTeX help. You must disclose the tools used, the purpose of use, and the affected parts of the work. The report's ideas, methodology, interpretation, and final wording must remain your own.

Q12: Can AI write my report?
No. A report that is mostly AI-generated, even if edited afterward, does not demonstrate your own learning and may receive a reduced grade. You may ask AI for feedback or language polishing, but the final report must be written, checked, and understood by you.

Q13: What happens if I submit late?
A penalty of **20% per day** is applied for late submissions. Reports submitted **more than 5 days late will NOT be accepted** and will receive a grade of zero.

Q14: What is the Personal Reflection & AI Use Disclosure Form?
It is a **separate document** submitted alongside your main report. It asks about your learning experience, challenges faced, tools used, AI-use disclosure, verification practices, and course feedback. Honest disclosure of permitted AI support will not be penalized by itself, but the form may be used to verify academic integrity and the student's own contribution.

Q15: Do I need to submit my code?
Yes. Submit your code via one of the following:

- GitHub repository link (preferred)
- Google Colab notebook link
- ZIP file uploaded to the Uzak system.

The code should be clean, documented, and reproducible. Include a README file or instructions on how to run it.

Q16: Can I use pre-built library functions (e.g., scikit-learn)?
Yes. You are encouraged to use established libraries. However, you must **demonstrate understanding** of the algorithms you use. Simply calling a function without explaining what it does, why you chose it, and how it works will result in a lower grade.

Q17: What if my results are not good?
That is perfectly fine. The project is about the **process**, not just the outcome. If your model accuracy is low or your patterns are not meaningful, **analyze why** in the Discussion section. Honest critical analysis of poor results is valued more than fabricated good results.

Q18: How should I handle class imbalance in my dataset?
If your dataset has class imbalance, you should:

- Document the imbalance in the Dataset Description section
- Apply appropriate techniques
- Use appropriate metrics (F1-score, precision-recall curve) instead of just accuracy
- Discuss the effect of imbalance handling in your results

Q19: Can I reuse a project from another course?
No. The project must be **original work** done specifically for this course. You may build

upon a topic you are interested in, but the implementation, analysis, and report must be new.

Q20: Who should I contact if I have questions?
Please use **Piazza**, as this allows everyone to benefit from the discussion.

12 Tips for Success

Tips for a Successful Project

1. **Start early.** Do not wait until the last week. Data collection and preprocessing often take longer than expected.
2. **Choose a topic you are genuinely interested in.** Curiosity drives better work.
3. **Understand your data before applying algorithms.** Spend adequate time on exploratory data analysis (EDA). Visualize distributions, check for missing values, understand correlations.
4. **Do not just run algorithms — understand them.** Be able to explain *why* you chose a particular method and *how* it works.
5. **Compare multiple approaches.** The best projects do not just apply one method — they compare several and analyze the differences.
6. **Visualize your results.** Good figures and plots communicate findings more effectively than tables of numbers alone.
7. **Write as you go.** Do not leave the report writing until the end. Document your methodology and results as you work.
8. **Be honest about limitations.** Every project has limitations. Acknowledging them shows maturity and understanding.
9. **Cite your sources.** Every claim, method, or idea that is not your own must be properly referenced.
10. **Proofread your report.** Check for spelling, grammar, formatting consistency, and logical flow before submission.
11. **Test your code.** Make sure your code runs from start to finish without errors. Include a README with instructions.
12. **Use version control.** Use Git/GitHub to track changes to your code. This also serves as evidence of your work progress.
13. **Ask for help early.** If you are stuck, do not wait — reach out via Piazza or email.

A Useful Links and Resources

A.1 Data Sources

- Kaggle Datasets: `https://www.kaggle.com/datasets`
- Kaggle Competitions: `https://www.kaggle.com/competitions`
- UCI Machine Learning Repository: `https://archive.ics.uci.edu`
- US Government Open Data: `https://www.data.gov`
- Google Dataset Search: `https://datasetsearch.research.google.com`
- AWS Open Data: `https://registry.opendata.aws`
- Papers With Code Datasets: `https://paperswithcode.com/datasets`

A.2 LaTeX Tools

- Overleaf: `https://www.overleaf.com`
- Prism (AI-assisted LaTeX editor): `https://prism.openai.com`
- IEEE Conference Template: `https://www.ieee.org/conferences/publishing/templates.html`
- LaTeX Tutorial: `https://www.overleaf.com/learn`

A.3 Python and Libraries

- Python: `https://www.python.org`
- scikit-learn Documentation: `https://scikit-learn.org/stable/`
- pandas Documentation: `https://pandas.pydata.org/docs/`
- mlxtend (Association Rules): `http://rasbt.github.io/mlxtend/`
- matplotlib: `https://matplotlib.org`
- seaborn: `https://seaborn.pydata.org`
- Google Colab: `https://colab.research.google.com`

A.4 Textbook

- J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Morgan Kaufmann, 2011.

Good luck with your project!

Remember: The goal is to learn, explore, and have fun with data mining.

Start early, ask questions, and enjoy the process.

— Dr. Hazm Talab
